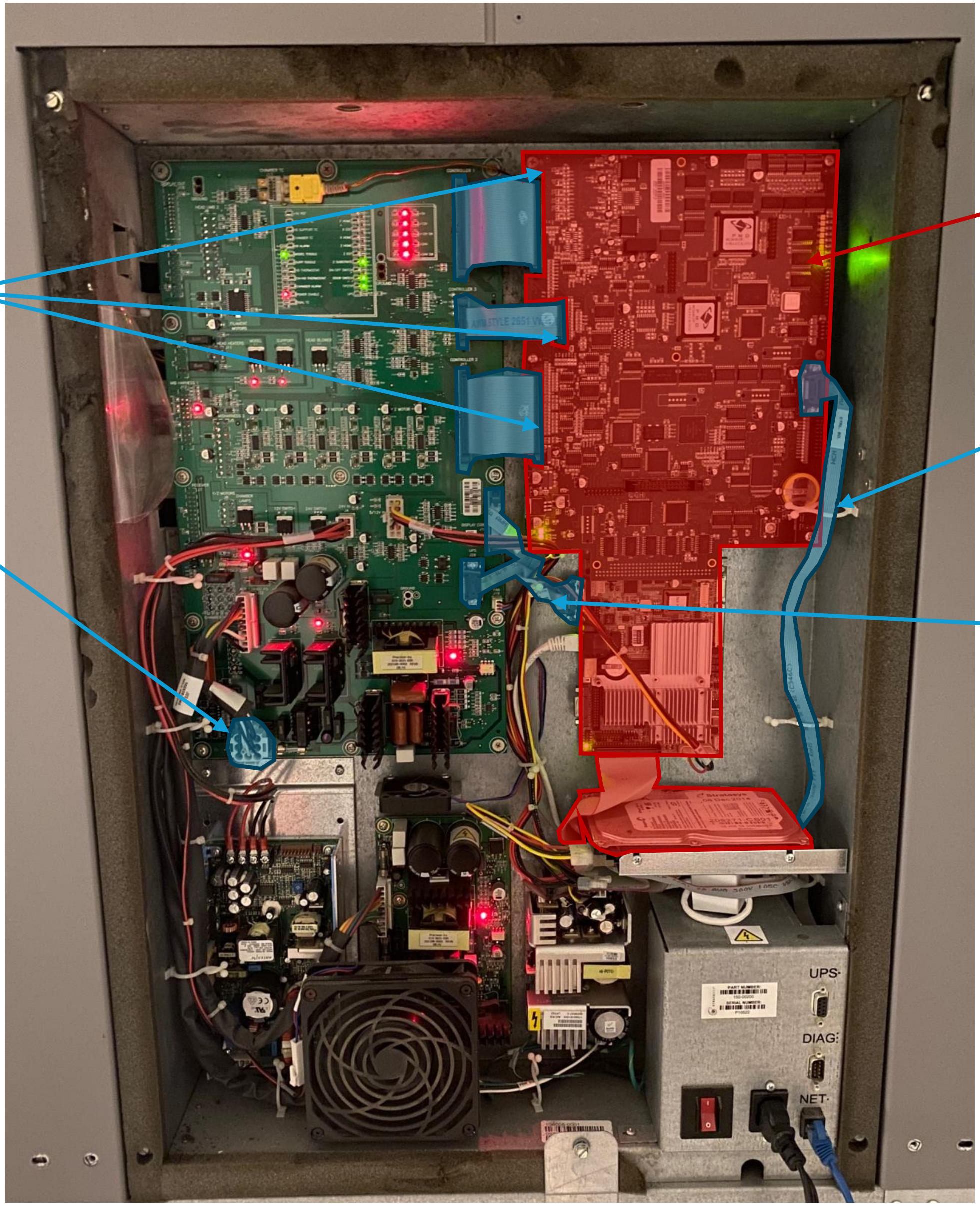
Disconnect 3x ribbon cables from the control board. Note the connectors are released via 2x lever locks on the short side of each board housing,

Disconnect power cable.

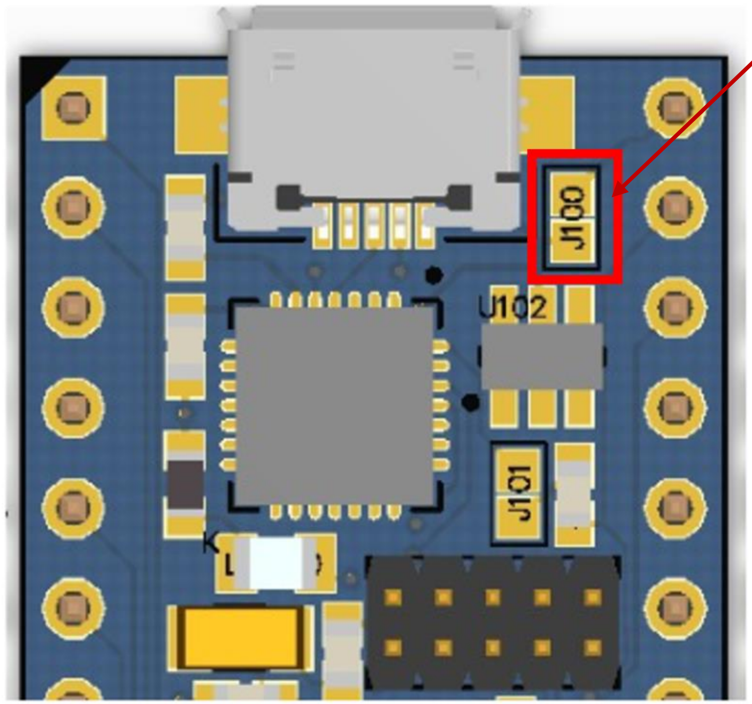


Remove and discard control board, SBC (single board computer), and hard drive. Discard screws and standoffs EXCEPT the screws used for the SBC, which are reused to mount the new 48V power supply.

Disconnect and tuck away/discard cable.

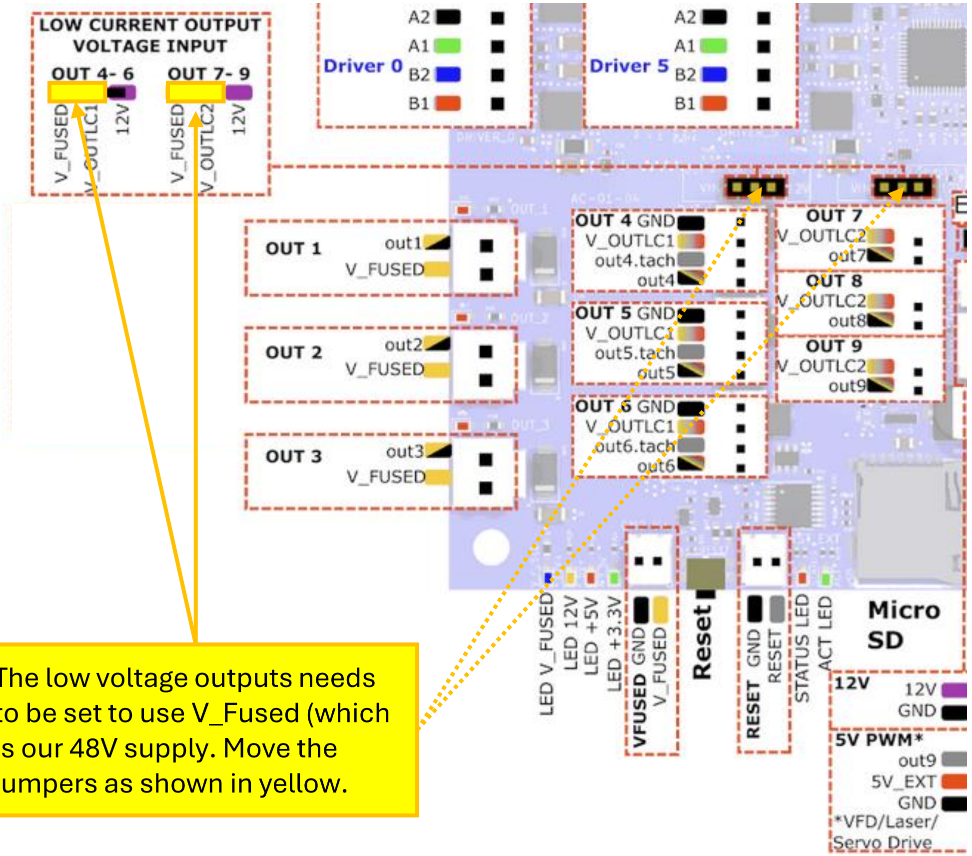
Disconnect and tuck away/discard cables.

Sammy-C21 Prep

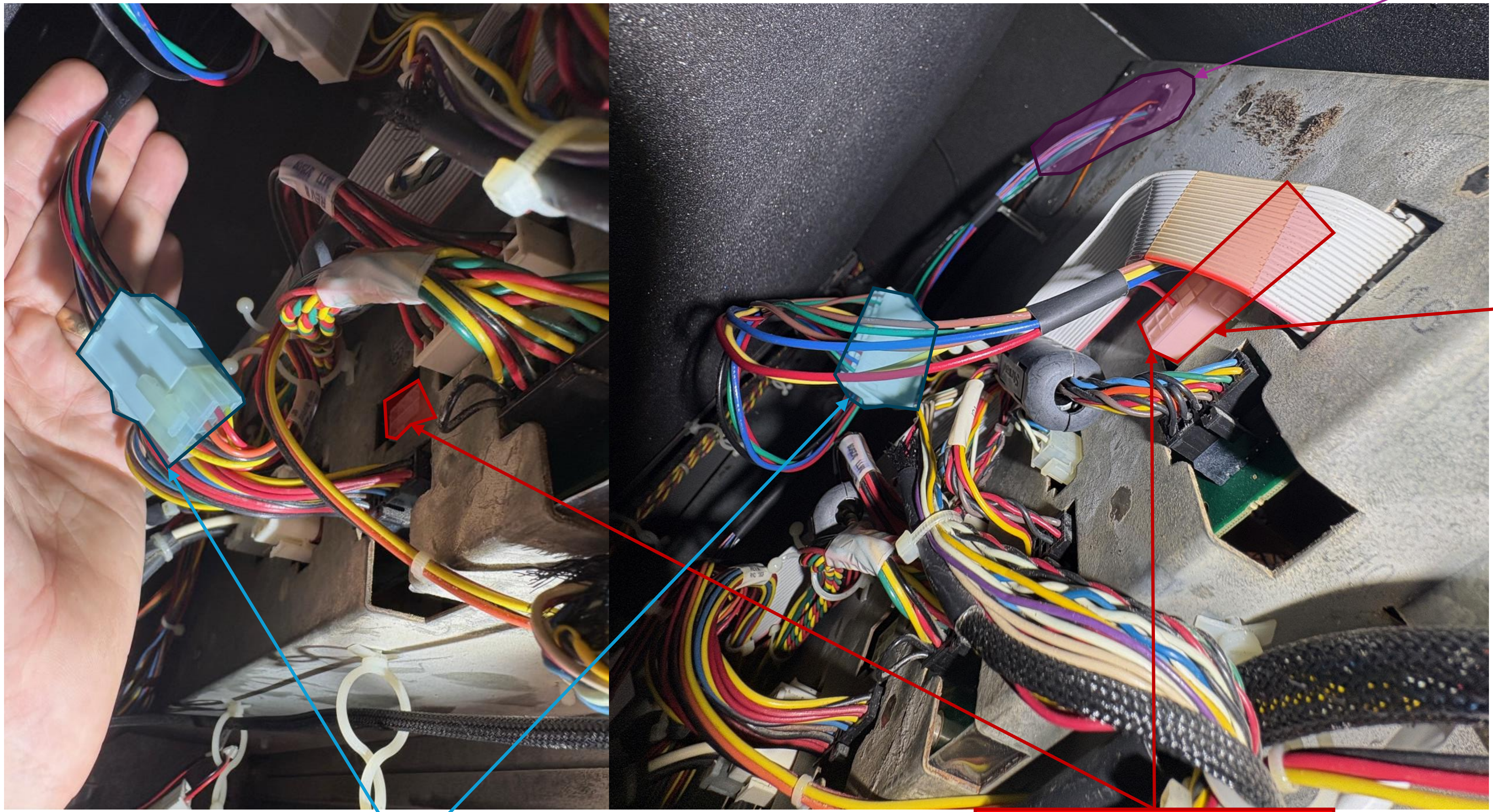


J100 needs to be unbridged. If bridged (soldered), use a soldering iron and soldering wick to unbridge (remove solder).

Duet 6HC Prep



The low voltage outputs needs to be set to use V_Fused (which is our 48V supply. Move the jumpers as shown in yellow.



1. Disconnect the XXx and XXx Mini-Fit Jr connectors from the PDB (power delivery board).

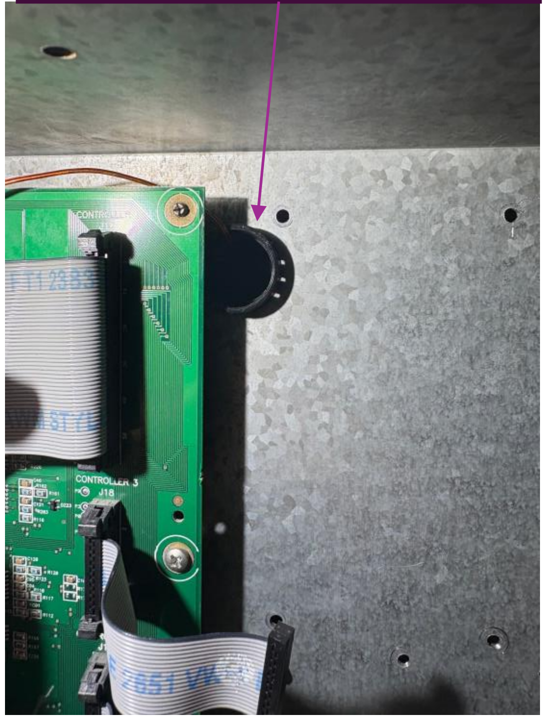
The 8 pin board connector will remain unused.

2. Connect the 8 pin and XX pin Mini-Fit Jr connectors from the new interface cable to the disconnected original connectors.

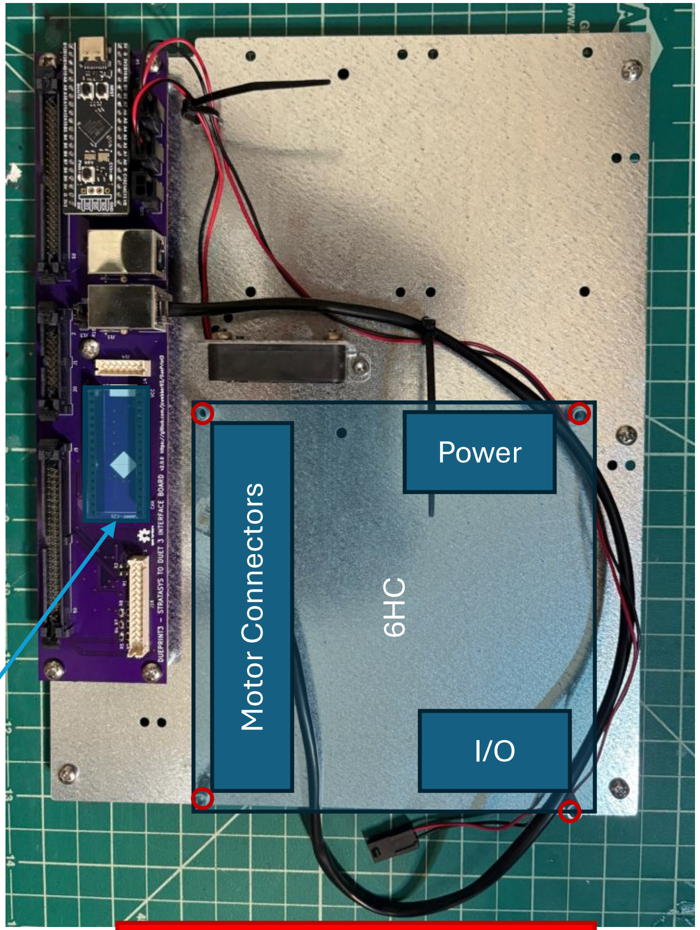
3. Connect the XX pin Mini-fit Jr connector from the interface cable to the PDB.

4. Pass the 3x Duet motor connectors through the feedthrough.

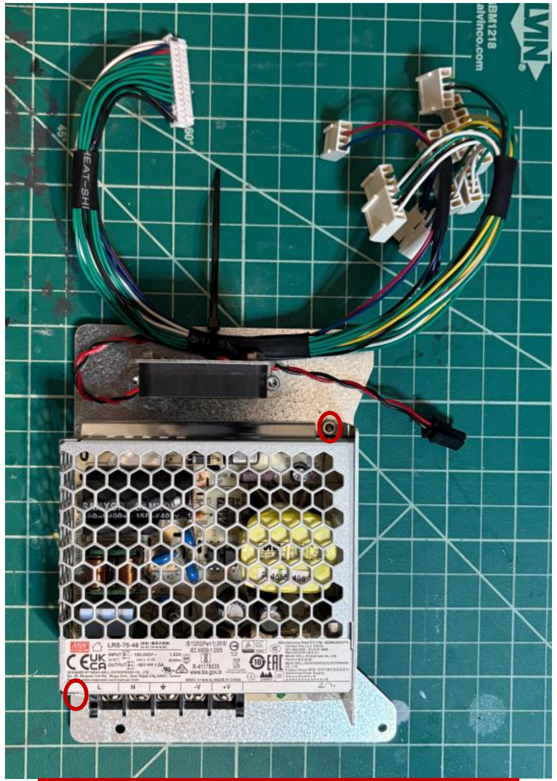
Pass the 3x Duet motor connectors through the feedthrough.



Install the Sammy-C21 as shown (USB pointing up!)



Mount the Duet 6HC to the panel with the 4x longer #6-32 screws.



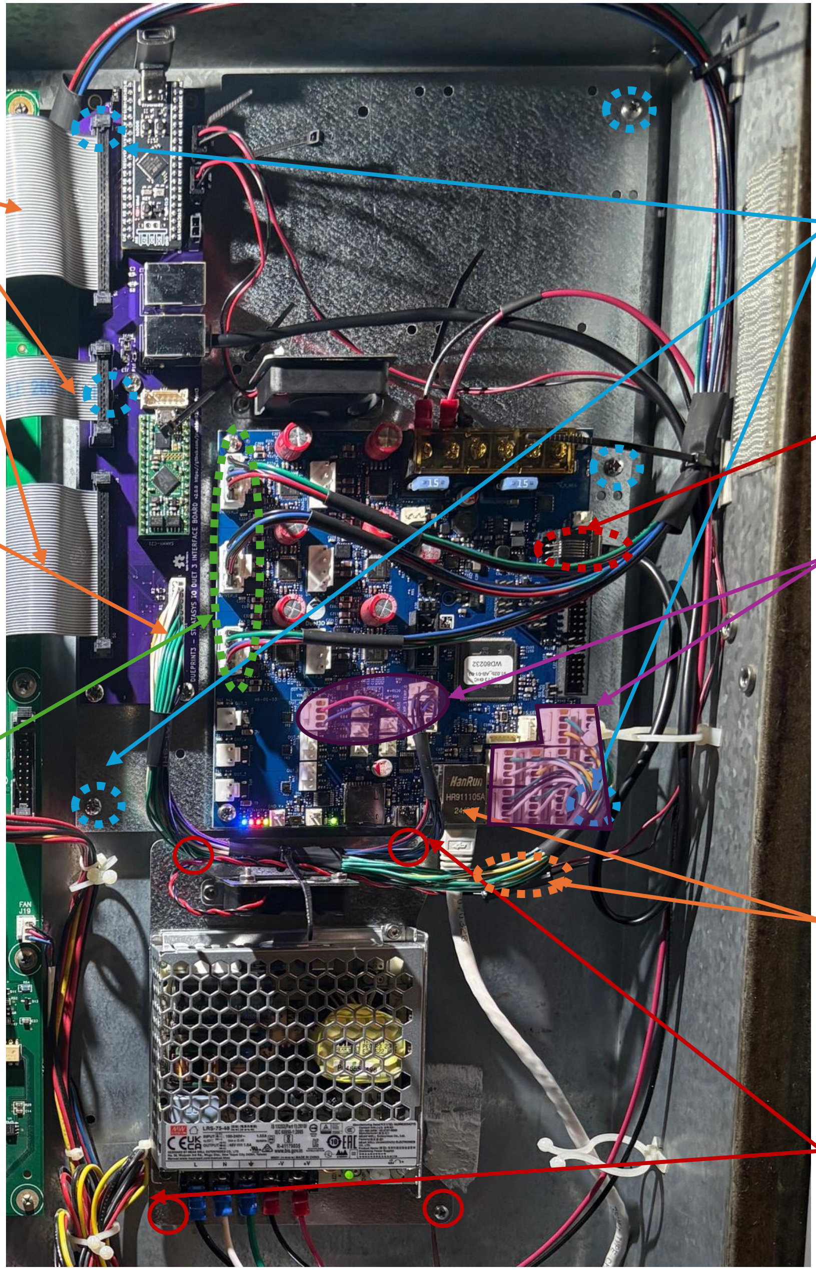
Mount the 48V PSU to the power supply panel via the 2x M3 screws.

Connect the 3x ribbon cables to the DuePrint3 Board.

Connect JST connector to header on DuePrint3.

Connect stepper motor connectors.

Driver 2 = Z
Driver 1 = Y
Driver 0 = X



Attach the new electronics panel into the machine via the 6x shorter #6-32 screws (which are fed from the inside area where the other cables are routed).

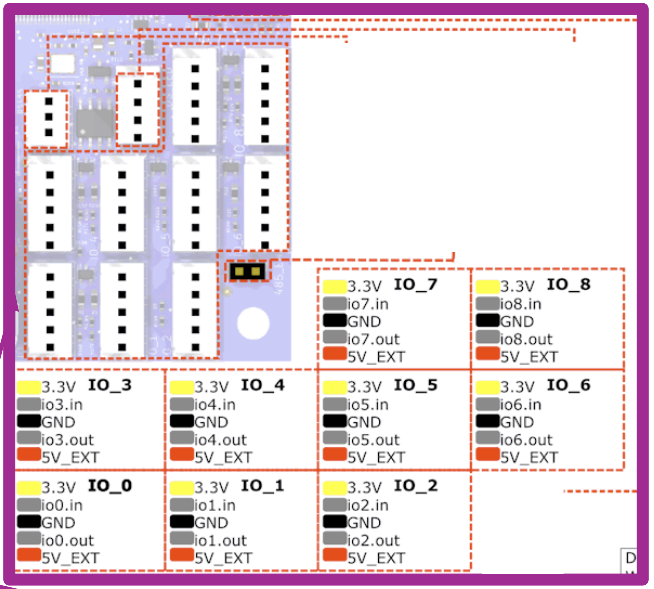
Connect RJ11 for CAN-FD

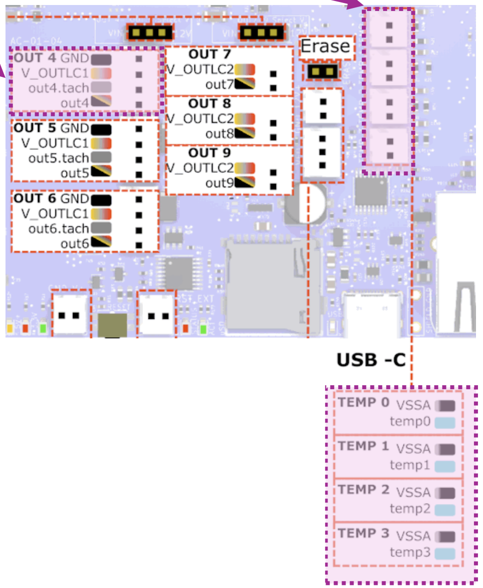
Connect I/O per markings on connectors

Plug ethernet cable into 6HC.

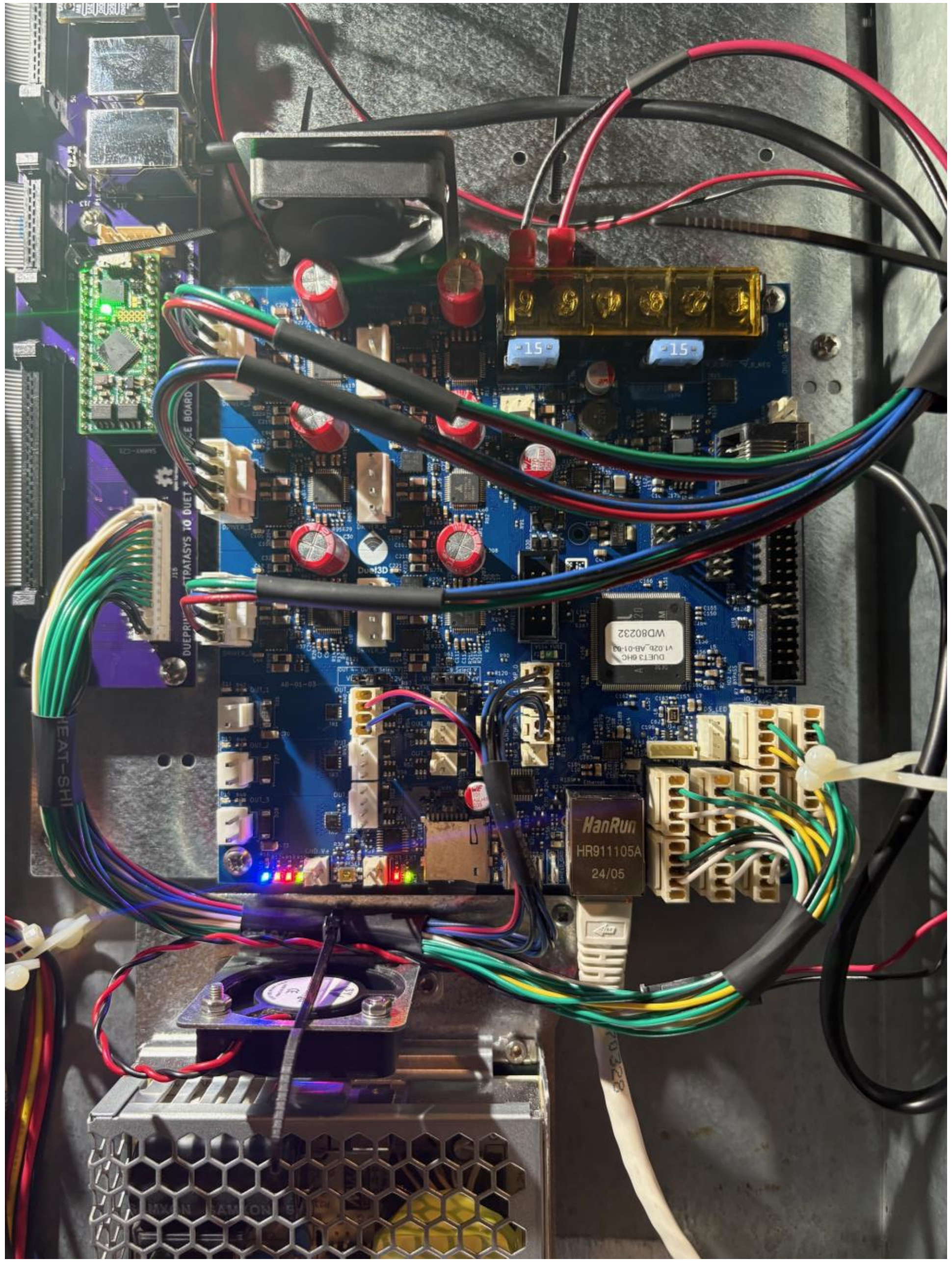
Connect 5V extension lead to 5V fan.

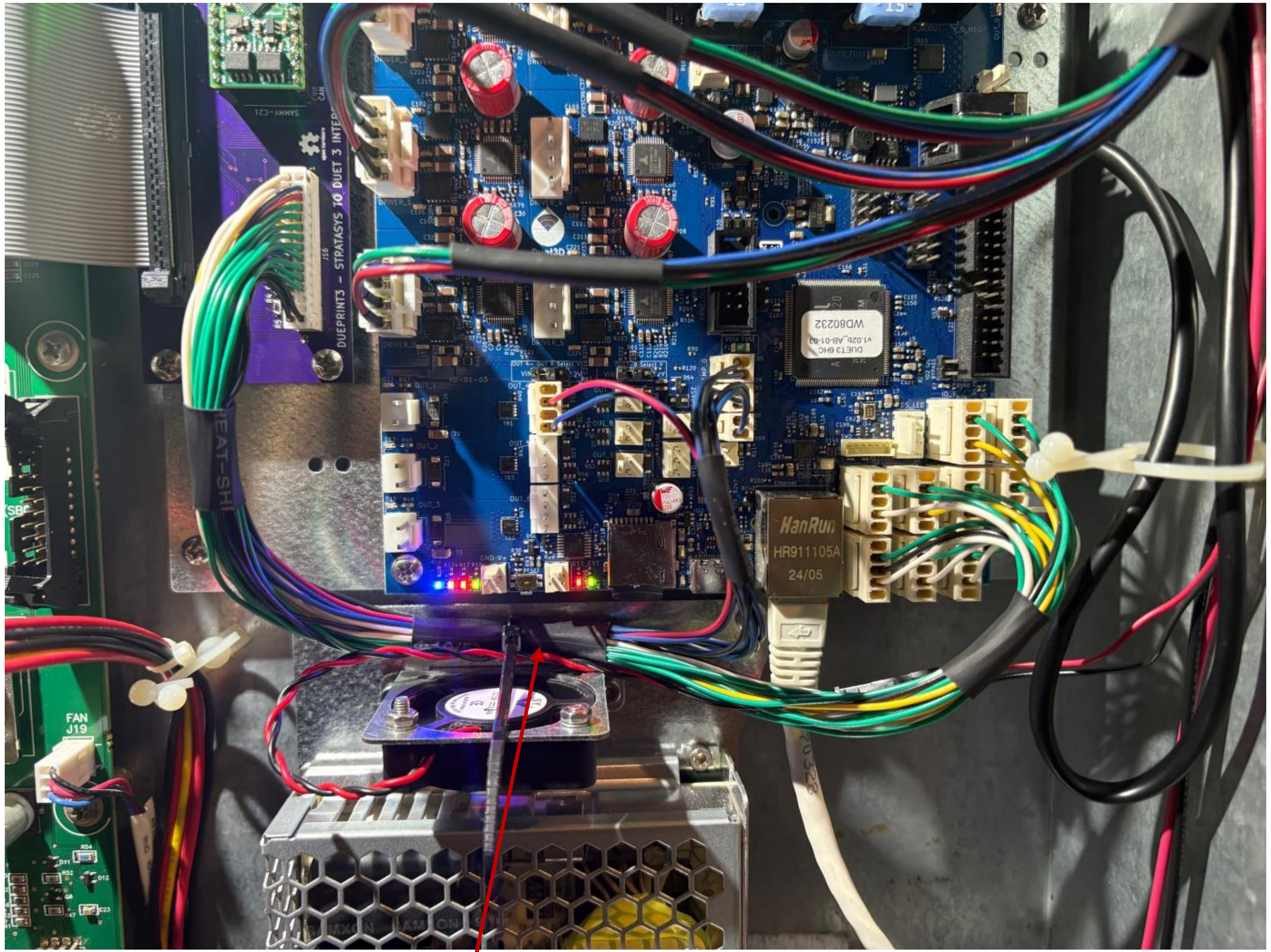
Attach the 48V power supply plate with the 4x screws previously used to mount the SBC.





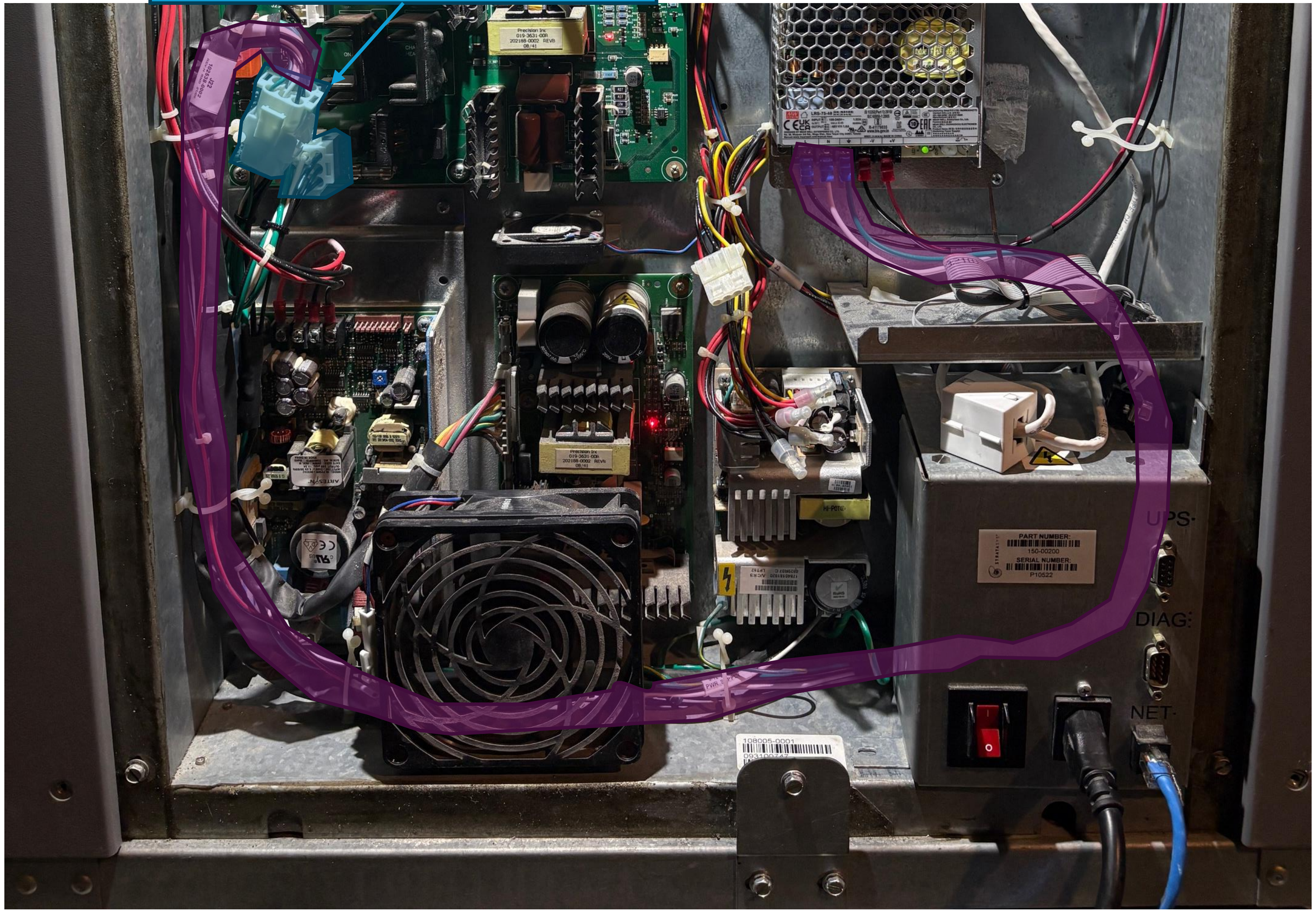
Additional images of connected Duet 6HC



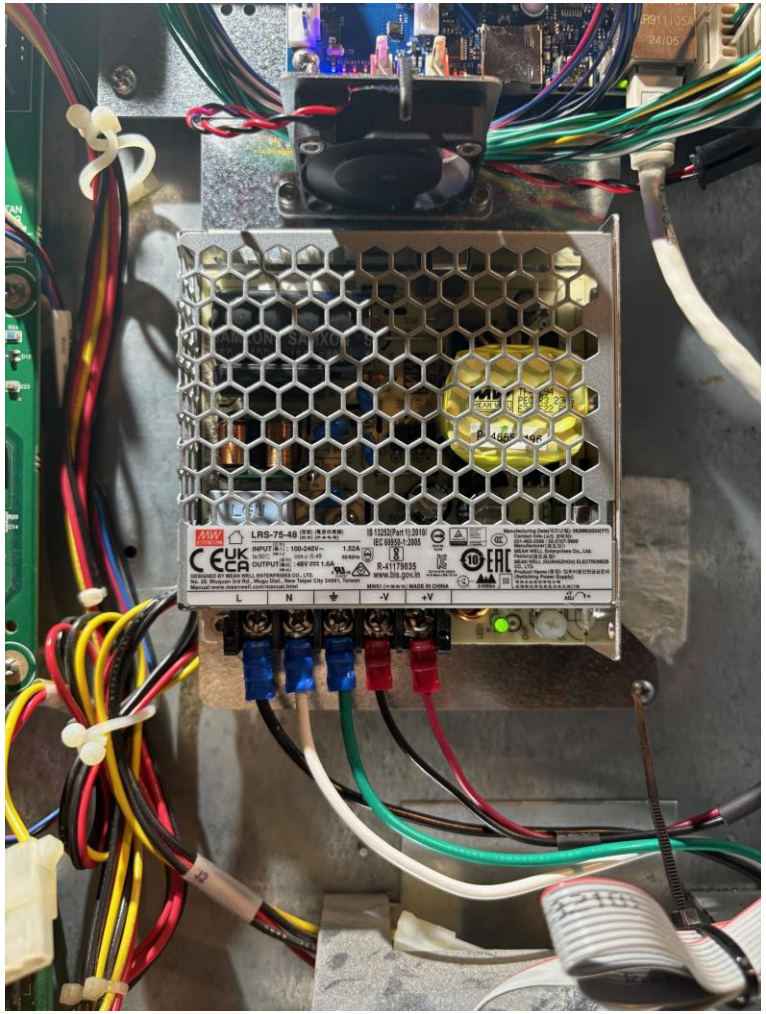


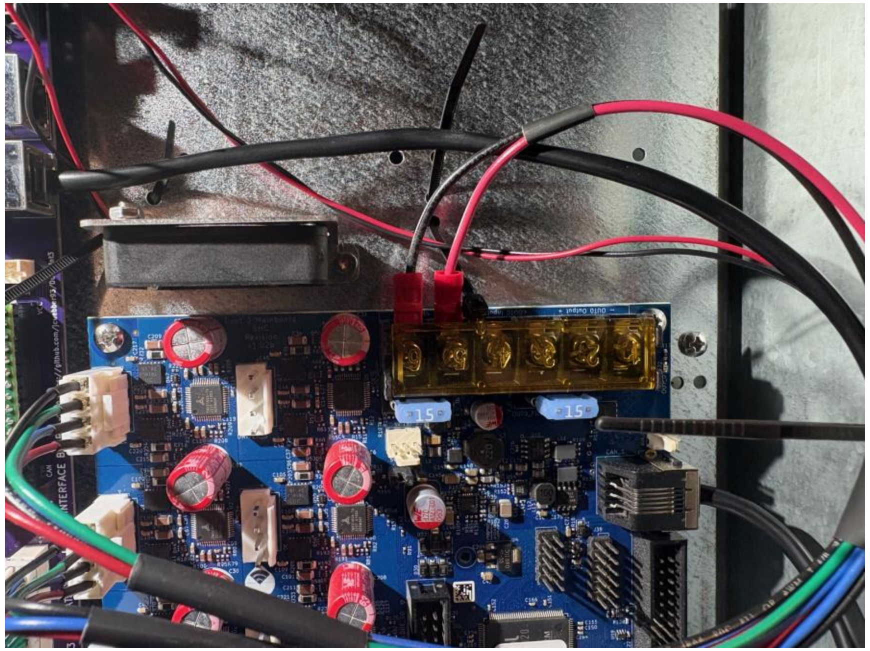
Ensure wires are not touching fan blades.

Connect new power harness to board.
Connect original power plug to new power harness.



Route power leads approximately as shown. You may wish to temporarily remove the power inlet module, which is retained by 3x screws.





AC

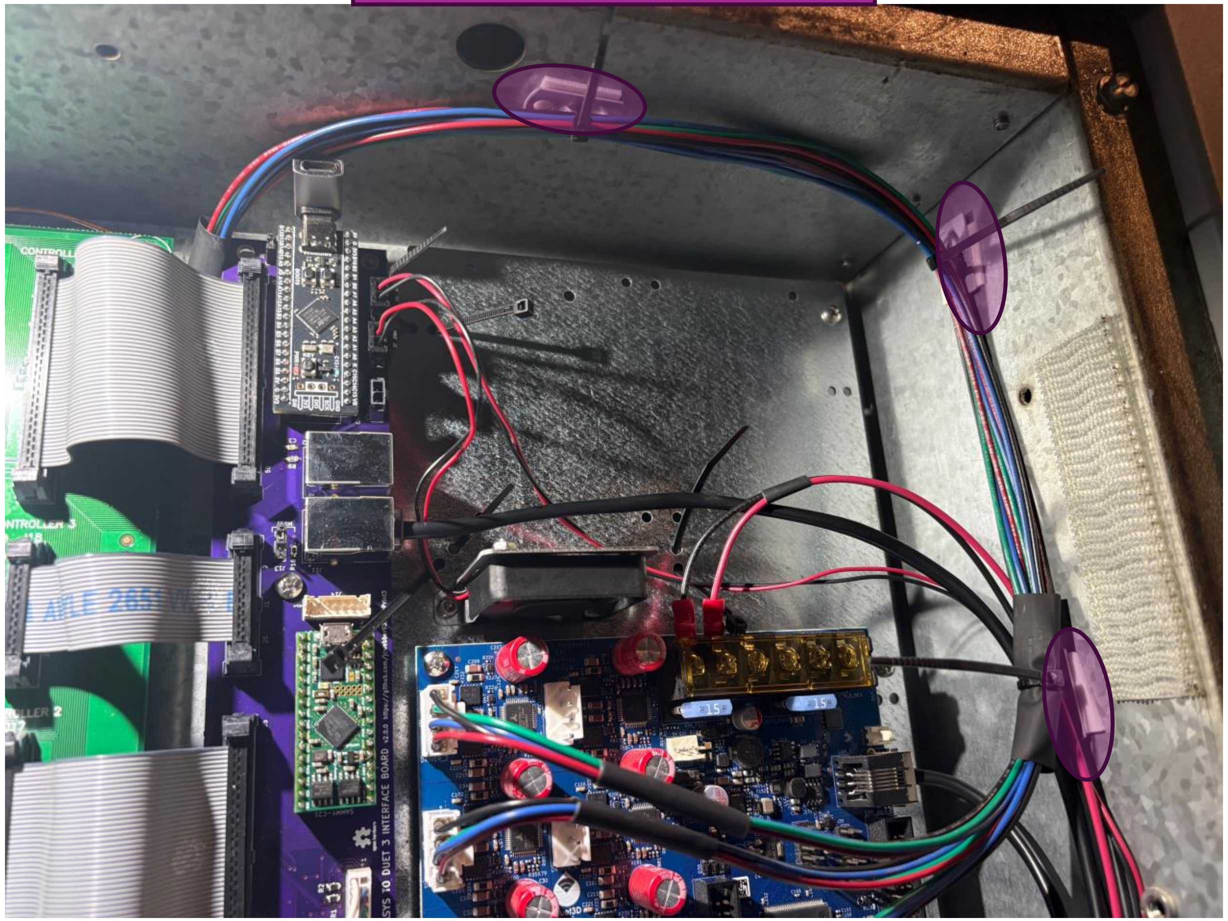
Live
Neutral
Ground

48VDC

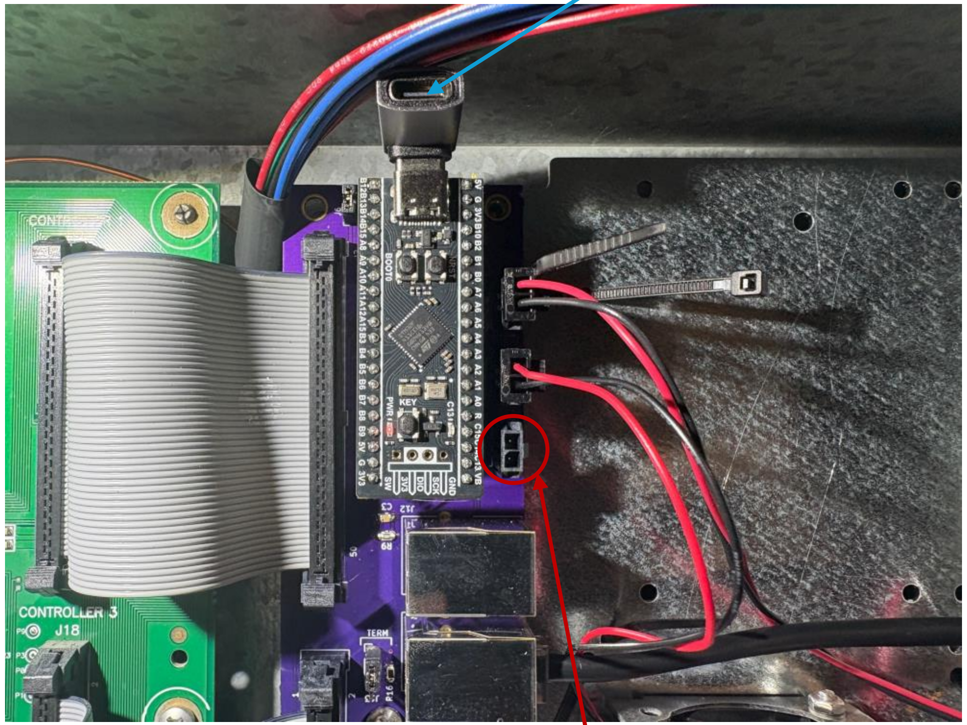
48V
Ground

Attach power terminals per the 6HC documents (for 48V) and 48V and AC power per your 48v power supply documents.

May wish to add tie down plates to aid with cable management.



Not included in your kit, but I used a 90° USB-C while testing the Blackpill.



Note this unpopulated connector is 12V. Do not accidentally connect a 5V fan to it!

This 12v connection is unused.